Answer **all** the questions in the spaces provided.

1 A ball is thrown horizontally from the top of a building, as shown in Fig. 1.1.

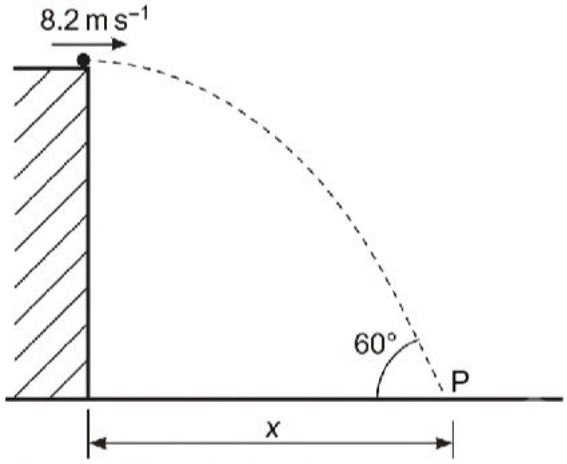


Fig. 1.1

The ball is thrown with a horizontal speed of 8.2 m s^{-1}. The side of the building is vertical. The ball hits the ground at point P at a distance x from the building and at an angle of 60° to the horizontal. Air resistance is negligible.

(a) For the ball at point P,

(i) show that the vertical component of its velocity is 14 m s^{-1},

$\tan\theta = \frac{v_y}{v_x}$ [1]

$v_y = v_x \tan\theta$

$= 8.2 \tan 60°$

$= 14.2$ [1]

$= 14 \text{ m s}^{-1}$

[2]

(ii) determine the vertical distance it has fallen,

$v_y^2 = u_y^2 + 2a_y s_y$

$14^2 = 0 + 2(9.81)s_y$ (Take downwards as +ve) [1]

$s_y = 9.99$

$= 10 \text{ m}$ (2 sf) [1]

vertical distance = ... m [2]

(iii) determine the horizontal distance x.

Let t be the time taken to fall through the height:

$$v_y = u_y + a_y t$$
$$14 = 0 + 9.81t$$
$$t = 1.43 \text{ s} \quad [1]$$

Then $s_x = u_x t$
$$= 8.2 \times 1.43$$
$$= 11.7$$
$$= 12 \text{ m (2 sf)} \quad [1]$$

distance x = ... m [2]

(b) State and explain the effect on your answer to the horizontal distance x if air resistance is not negligible.

Air drag causes the horizontal velocity to continually decrease throughout the motion. [1]

Because the average horizontal speed during the ball's flight is lower, the ball covers less horizontal distance in a given time [1] [2]

[Total: 8]